

Augmented Reality Game for Brand Recognition and Recall

AR for Brand Recognition and Recall

Carlos, Arce-Lopera
Faculty of Engineering, Universidad
Icesi
caarce@icesi.edu.co

Rafael, Ospina
Faculty of Engineering, Universidad
Icesi
rafael.theoc@gmail.com

Daniela, Victoria
Faculty of Engineering, Universidad
Icesi
danivictoria93@gmail.com

ABSTRACT

Augmented Reality (AR) technology offers the opportunity to change the customer experience in the retail context. For example, AR applications can be used as a bridge to close the gap between physical and online shops, mixing the best characteristics of both. Different researchers have measured the quality of the user's experience through AR. However, here we focused not only on user experience but also on the impact of AR on brand recall and recognition. We analyzed the impact on brand recall and recognition of an AR application by comparing three contexts: a traditional context where participants look at three products, a focused context where participants have only one product to look at, and a technology-enhanced context where participants interacted with the product using the designed AR experience. Experimental results revealed that AR improved brand recall and recognition by creating an engaging environment. Moreover, participants in the AR context recalled additional information about the product. Also, the AR experience motivated them to participate in an omnichannel activity based on a QR interaction. These results show the influence of well-designed AR experiences on brand recall and recognition opening the possibility for new technology in retail environments.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Customer experience, Augmented Reality, Brand Recall, Brand Recognition, Technology in Retail, Omnichannel, QR

ACM Reference Format:

Carlos, Arce-Lopera, Rafael, Ospina, and Daniela, Victoria. 2021. Augmented Reality Game for Brand Recognition and Recall: AR for Brand Recognition and Recall. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3411763.3451792>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451792>

1 INTRODUCTION

In the last decades, enhancing user experience has gained importance in different contexts. In particular, in commercial settings, the application of different types of technology to entertain customers has been used to improve the shopping experience, and promote products and services in an attractive way [6, 7]. The increasing focus on customer experience arises because customers now interact with the products through a myriad of touchpoints in multiple channels and media. This results in more complex customer journeys for retailers [8], where brands need to build remarkable experiences to improve brand recognition and recall. Technology is a tool that can create an attractive and personalized environment, making the shopping experience engaging and memorable [2]. However, while enhancing customer experience significantly affects the purchase intention, it is important that customers perceive an added value during the customer journey and purchase process to improve their level of satisfaction [10].

By exploring innovative methods to enhance the customer experience and satisfaction, the physical retail environment is reinventing itself and adopting emerging technology into its entire shopping process [11]. In this regard, retailers should pay attention to adopt smart technologies that are user-friendly. Currently, immersive technologies, such as Augmented Reality (AR) and Virtual Reality (VR), are rapidly evolving and are increasingly being adopted in the retail context [13]. For instance, both VR and AR are gaining recognition by marketers because of their potential to create a differentiated and personalized customer retail experience [3].

AR is a technology that enhances the user's sensory perception of the real world with a contextual layer of information. By compensating for the lack of product information or the inability to handle the product, AR creates a three-dimensional augmented experience driven by perceived benefit and value [12]. AR applications for the retail environment commonly provide 3D product information. For example, different colors and styles can be superposed like magic mirrors which enhances users' perception of reality [2]. Also, mixed AR provides the user with enriched product information gained from a physical store as well as an online store. While these applications use technology, they need to be designed to focus on creating a memorable experience to increase the perceived value for customers. The issue remains on how to carefully design and select the stimuli as not to generate an overwhelming experience.

On the other hand, gamification can also improve customer involvement and engagement [8]. In the retail environment, gamification is defined as the process of integrating game-thinking and game-mechanics to engage the customer in the non-gaming context of shopping [4]. Moreover, customer engagement through a game-like experience shows the potential benefits that derive from the



Figure 1: AR prototype interaction

introduction of gamification in the Fast-Moving Consumer Goods sector [8]. Therefore, if brand-attributed benefits are effectively communicated to the public, it can contribute towards brand differentiation, improve customers' attitude and evaluation, and provide better resistance to price-induced competition attacks.

2 METHODS

AR applications are commonly used in the retail context due to their low cost of implementation and easiness for customers to access information via their smartphones. To test for the influence of immersive technology on brand recognition and recall, an AR application with different types of interactions was developed.

2.1 Product Description

To use as a target product, we selected a half-pound package of single-origin organic coffee from Cauca, Colombia. The coffee is produced by a small company founded in 2018 called "Café Giosma". The brand is relatively new in the market which was convenient for the purpose of this research.

2.2 Prototype Description

In order to develop the interaction, an AR environment was built based on the game engine software Unity3D. One of the advantages of using Unity3D is the possibility to publish projects on any mobile platform. The mobile app was designed to show augmented reality effects while scanning different points in a coffee bag package (see

Figure 1). In Figure 1, section 0 shows a person using the app, from section 1 to 9, screenshots from the application are shown.

First, in front of the bag, a menu that described all the possible AR interactions with the bag was shown (see Figure 1.1). This allowed the user to watch an image sequence concerning the coffee harvest and production (Figure 1.2 and 1.3). Secondly, pointing to a label with the camera, the AR app scanned a figure of a man and played a video featuring a story told by the coffee farmer (see Figure 1.4). Finally, in the back, a game interaction was designed. First, a game tutorial video (see Figure 1.5) was played followed by the display of the game interface. For the game, an upper layer showed a text corresponding to one of three ingredients (see Figure 1.6, 1.7, 1.8). Then, the user had to search for the ingredient in the package using the mobile camera. Once the ingredient was identified, a score changed and the task continued with another ingredient. After 1 minute, the final score was shown on a dashboard (see Figure 1.9).

On the other hand, a coffee sample with a QR (See Figure 2) was delivered to every participant. The QR code redirected to a website with general information about coffee. Moreover, interested users could fill a web form to participate in a random draw for a pound of coffee.

2.3 Participants

A total of 90 participants were recruited from the Universidad Icesi in Colombia. All participants were randomly selected from different University faculties. 55 males and 35 females participated. Their



Figure 2: Coffee samples with QR.

age was from 18 to 32 years old with an average of 20.6 $\pm$ 2.9 years old. All participants were regular coffee consumers and were avid and experienced mobile users.

2.4 Experimental Design

A between-subject experimental design was adopted. Subjects were organized into three test groups: Base, Control, and AR. Subjects in the Base group participated in a simulation of a common retail environment where customers explore products in front of shelves. Participants were asked to visually inspect three different coffee brands for one minute each, the target brand, and two more brands. In the Control group, participants were asked to spend three minutes watching one coffee package from one of the three coffee brands and selected randomly to avoid any selection bias. The act of visually inspecting a product for longer or on different occasions makes it more likely that this product will be bought or remembered [2]. In the AR group, participants inspected a coffee package from the target brand using the AR mobile application for a maximum of three minutes. Furthermore, videos were recorded during all the experimental conditions to analyze user reactions and expressions.

After each task in all groups was performed, two surveys were given to participants. First, the User Engagement Scale Short Form (UES-SF) was given immediately at the end of each session. The UES-SF measures engagement using four dimensions: FA (Focus Attention), PU (Perceived Usability), AE (Aesthetic Attractiveness),

and RW (Reward) [9]. Secondly, a survey was given 24 hours after the experimental session. The survey included questions to measure brand recognition and recall. Brand recognition is how quickly a consumer recognizes and discriminates the brand when any of its elements is shown, e.g., logo, slogan [5]. On the other hand, brand recall may happen when the consumer recognizes or identifies a brand using only information from memory or when the consumer is able to associate the brand with given sensory cues [1]. Brand recall was tested by asking subjects to remember the brand name without any clues. Then, brand recognition was assessed by asking participants to choose the name among four options. Additionally, we asked for specific information on the target brand to test for the recall of extra information.

To summarize, data was collected from four different sources: the UES short form, the survey 24 hours after the test, the interaction with the QR code, and the video analysis of the AR interactions. For the QR interaction, the number of participants of each group and the elapsed time between the participation and the QR register was recorded. The videos were analyzed individually, and participant reactions were quantified manually by experts. Participants' reactions were extracted by counting the number of times participants viewed the brand name. Then, surprise behavior was quantified by identifying the number of smiles and fascination reactions showed by participants during the interaction. Finally, frustration behavior was identified for each participant that showed anxiety for finishing the task or when they felt lost.

3 RESULTS

Table 1 summarizes the behavior of the participants in each group, the third column shows the number of participants who answered the survey administered 24 hours after the experiment while the fourth column shows the number of participants that participated in the omnichannel activity by using the QR code. In total, 47% of all participants answered the survey administered 24 hours after the experiment but only 11% participated in the omnichannel experience.

Table 2 presents the results for the brand recall and recognition in the next 24 hours test. The third and fourth columns show the number of participants who could correctly answer the questions. The last column shows the number of participants that could correctly recall additional information from the product.

Table 3 shows the results of the video analysis. The third column presents the total of times participants watched the brand name. The last column is the total of participants that showed any signs of frustration during the experience.

Finally, Figure 3 shows the results for the UES short form. Grey bars represent the result for each group. The AR group results are

Table 1: Participants in 24 hours' test and omnichannel activity

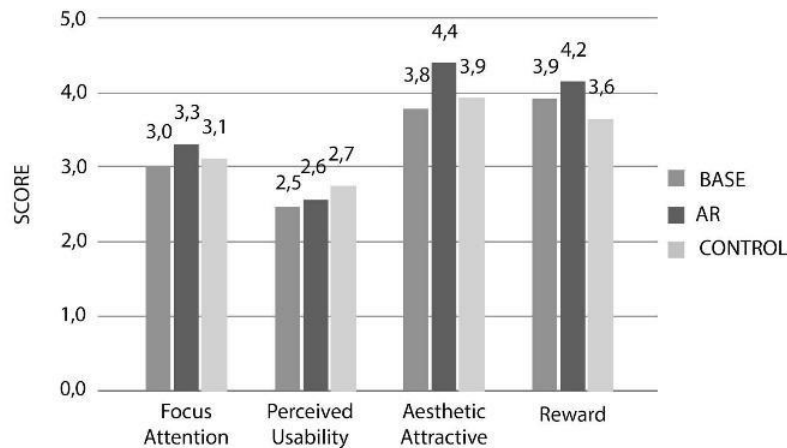
Group	No. of Participants	24h	QR registrations
Base	30	8	0
AR	30	16	6
Control	30	18	4
Total	90	42	10

Table 2: Brand recall and recognition in 24 hours' test

Group	Total 24h test	Correct Recall	Correct Recognition	Correct Additional Info
Base	8	0	3	2
AR	16	4	11	10
Control	18	6	11	13
Total	42	10	25	25

Table 3: Behavior identified based on video analysis.

Group	No. of Participants	No. Brand Views	No. Surprise Reactions	Frustration
Base	30	96	4	11
AR	30	235	130	7
Control	30	251	4	20

**Figure 3: UES Short Form results.**

indicated by the darkest bar while the Control group results are displayed in the lightest grey bar.

4 DISCUSSION

To validate if the use of AR technology increases the interest to perform an omnichannel activity, coffee samples with a QR were distributed to all participants. As shown in Table 1, 20% of subjects in the AR group and 13% of subjects in the Control group participated in the QR omnichannel experience. In contrast, none of the participants of the Base group used the QR. Thus, when compared to a more traditional customer experience (Base group), the designed retail experience based on AR increased in 20% the interest for omnichannel participation. Furthermore, the response times for the omnichannel activity, i.e., the elapsed time from getting the QR code until accessing it, for the Control group was up to 8 hours after the experiment. In contrast, for the AR group, participants registered using the QR code up to 7 days after the experiment. This result may indicate that the AR experience can keep customers motivated to participate in an omnichannel activity for a longer period of time.

Table 2 shows results on whether the AR interaction improved the brand recall in comparison to a usual retail experience (Base group). In the Base group, none of the participants recalled the brand name. In contrast, 25% of participants in the AR condition responded correctly. With a confidence of 70% ($p = 0.262 < 0.3$), a Fisher's Exact Test showed a statistically significant relationship between recalling the brand and the type of interaction (Base vs AR). Besides, when testing for brand recognition, only 37,5% of the Base group chose the correct brand name, against 68.8% of correct responses for the AR group, revealing a significant difference between the AR and Base groups, with a confidence of 70% ($p = 0.204 < 0.3$). On the other hand, for the Control group, 33% of participants recalled the brand name and 61% selected the correct answer in the recognition task. However, Fisher's exact test showed that there was no difference between doing the AR interaction or the Control interaction for brand recall ($p = 0.715$) or recognition ($p = 0.729$). This result may indicate that both interactions (AR vs Control) had similar effects on brand recognition and recall.

To analyze the relationship between the brand recall and the times a participant sees the brand name, data from all the

participants were organized into two groups based on their ability to recall the brand. There was a difference with a confidence of 80% ($t(15) = 1.43$, $p\text{-value} = 0.17 < 0.2$), between the people who could recall the brand according to the number of times that the person watched the brand name. Participants who recalled the brand name have a larger mean of 12 views in comparison with the 9 views for the rest of the participants. Moreover, the AR mobile application was the scenario with a major influence with a median of 14 views, the second one was the control group with 10 views. The base group reported only 2.5 views. Moreover, a significant difference with 80% of confidence was found for the effect of the number of views of the brand name and its recognition ($t(38) = 1.35$, $p\text{-value} = 0.1855 < 0.8$). Participants who could recognize the brand in the multiple-choice option watched on average of 10.8 times the brand name against 8.6 times for those who could not recognize the brand.

To test if AR technology can help to recall extra information from the target brand, we asked subjects if they could remember additional information of the product. When comparing the AR and Base groups against the recall of extra information using the Fisher's Exact Test, a statistically significant relationship with a confidence of 80% ($p = 0.193 < 0.2$) was found, meaning that the type of interaction influences the assertiveness to recall extra information. Furthermore, the AR group had better recall of related information with 62.5% against 25.0% for the Base group (see Table 2). No difference was found between the AR group and the Control group, i.e., the AR experience is similar to watching the product for three minutes. However, AR allows a significantly less frustrating environment (see Table 3) with 7 frustrated people in comparison to 20 for the Control group ($p < 0.01$).

Finally, data analysis of the results of the UES-SF (see Figure 3) revealed that there was a significant difference for all four dimensions of engagement. Analysis of variance yielded significant differences among all groups in perceived usability ($t(2) = 2.201$, $p < 0.15$), focus attention ($t(2) = 2.381$, $p\text{-value} < 0.1$), aesthetic perception ($t(2) = 3.6$, $p < 0.05$) and reward perception ($t(2) = 5.0$, $p < 0.01$). Moreover, Tukey HSD showed a significant difference between participants in the AR group and the other two groups. These results suggest that the AR interaction offered more engagement to participants.

5 CONCLUSIONS

Using AR in retail environments revealed an increase in customer engagement with the shopping experience and an enhancement in brand recognition and recall. The effect on brand recall and recognition is similar to forcing the customer to focus on the product for three minutes. However, AR was more engaging and less frustrating than the other two experimental conditions. Indeed, according to our experimental results, people that interacted with the AR condition were engaged by the gamification experiences and enjoyed

the interactions. AR is a display technology that can increase brand recall in consumers by simply increasing brand visual exposure throughout the shopping experience. Therefore, the inclusion of AR technologies in a traditional retail context can serve as a tool to support marketing efforts that deliver and guide different perceived value to the customer. Future research should focus on continuing to investigate what design decisions in AR applications make the retail experience more engaging and somehow memorable.

REFERENCES

- [1] Chris Baumann, Hamin Hamin, and Amy Chong. 2015. The role of brand exposure and experience on brand recall—Product durables vis-à-vis FMCG. *J. Retail. Consum. Serv.* 23, (March 2015), 21–31. DOI: <https://doi.org/10.1016/j.jretconser.2014.11.003>
- [2] Marta Blazquez Cano, Patsy Perry, Rachel Ashman, and Kathryn Waite. 2017. The influence of image interactivity upon user engagement when using mobile touch screens. *Comput. Hum. Behav.* 77, (December 2017), 406–412. DOI: <https://doi.org/10.1016/j.chb.2017.03.042>
- [3] Francesca Bonetti, Gary Warnaby, and Lee Quinn. 2018. Augmented Reality and Virtual Reality in Physical and Online Retailing: A Review, Synthesis and Research Agenda. In *Augmented Reality and Virtual Reality: Empowering Human, Place and Business*, Timothy Jung and M. Claudia tom Dieck (eds.). Springer International Publishing, Cham, 119–132. DOI: https://doi.org/10.1007/978-3-319-64027-3_9
- [4] Sebastian Deterding, Dan Dixon, Rilla Khaled, and Lennart Nacke. 2011. From Game Design Elements to Gamefulness: Defining “Gamification.” In *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments (MindTrek '11)*, ACM, New York, NY, USA, 9–15. DOI: <https://doi.org/10.1145/2181037.2181040>
- [5] Mehreen Khurram, Faisal Qadeer, and Muhammad Sheeraz. 2018. *The Role of Brand Recall, Brand Recognition and Price Consciousness in Understanding Actual Purchase*. Social Science Research Network, Rochester, NY. Retrieved January 10, 2021 from <https://papers.ssrn.com/abstract=3215875>
- [6] Youn Kyung Kim. 2001. Experiential retailing: an interdisciplinary approach to success in domestic and international retailing. *J. Retail. Consum. Serv.* 8, 5 (September 2001), 287–289. DOI: [https://doi.org/10.1016/S0969-6989\(00\)00034-5](https://doi.org/10.1016/S0969-6989(00)00034-5)
- [7] Robert V Kozinets, John F Sherry, Benet DeBerry-Spence, Adam Duhachek, Krittinee Nuttavuthisit, and Diana Storm. 2002. Themed flagship brand stores in the new millennium: theory, practice, prospects. *J. Retail.* 78, 1 (March 2002), 17–29. DOI: [https://doi.org/10.1016/S0022-4359\(01\)00063-X](https://doi.org/10.1016/S0022-4359(01)00063-X)
- [8] Stavros Lounis, Xanthippi Neratzouli, and Katerina Pramatar. 2013. Can Gamification Increase Consumer Engagement? A Qualitative Approach on a Green Case. In *Collaborative, Trusted and Privacy-Aware e/m-Services (IFIP Advances in Information and Communication Technology)*, Springer Berlin Heidelberg, 200–212.
- [9] Heather L. O'Brien, Paul Cairns, and Mark Hall. 2018. A practical approach to measuring user engagement with the refined user engagement scale (UES) and new UES short form. *Int. J. Hum.-Comput. Stud.* 112, (April 2018), 28–39. DOI: <https://doi.org/10.1016/j.ijhcs.2018.01.004>
- [10] Eleonora Pantano and Giuseppe Naccarato. 2010. Entertainment in retailing: The influences of advanced technologies. *New Technol. Retail. Trends Dir.* 17, 3 (May 2010), 200–204. DOI: <https://doi.org/10.1016/j.jretconser.2010.03.010>
- [11] Eleonora Pantano, Constantinos-Vasilios Priporas, Stefano Sorace, and Gianpaolo Iazzolino. 2017. Does innovation-orientation lead to retail industry growth? Empirical evidence from patent analysis. *J. Retail. Consum. Serv.* 34, (January 2017), 88–94. DOI: <https://doi.org/10.1016/j.jretconser.2016.10.001>
- [12] Atieh Poushneh and Arturo Z. Vasquez-Parraga. 2017. Discernible impact of augmented reality on retail customer's experience, satisfaction and willingness to buy. *J. Retail. Consum. Serv.* 34, (January 2017), 229–234. DOI: <https://doi.org/10.1016/j.jretconser.2016.10.005>
- [13] P. Rosedale. 2017. Virtual Reality: The Next Disruptor: A new kind of worldwide communication. *IEEE Consum. Electron. Mag.* 6, 1 (January 2017), 48–50. DOI: <https://doi.org/10.1109/MCE.2016.2614416>